ASIR

HARDWARE FUNDAMENTALS

30/09/2026

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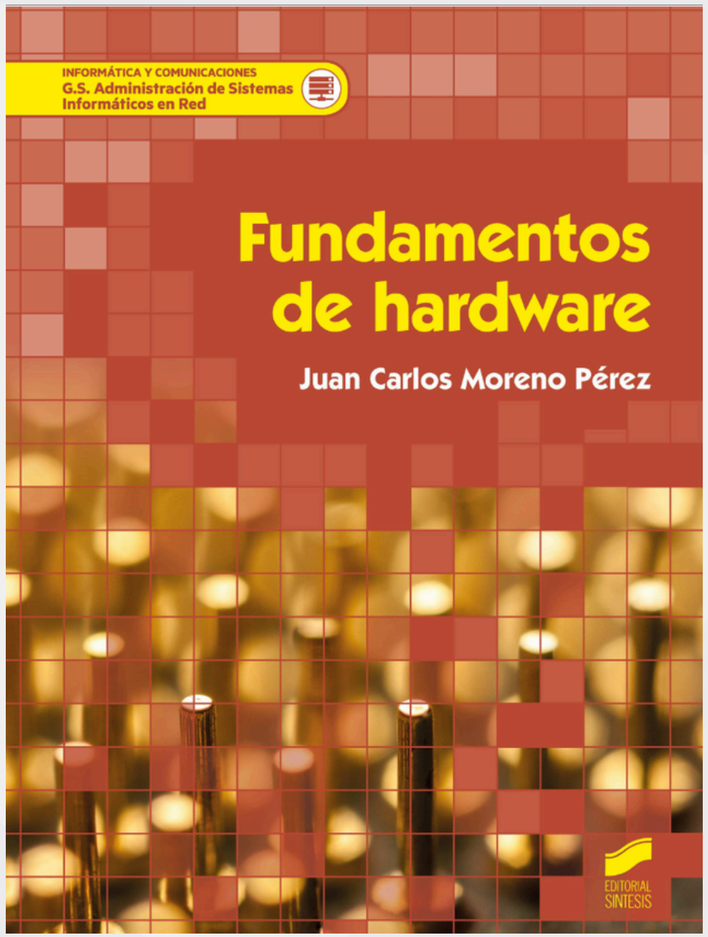
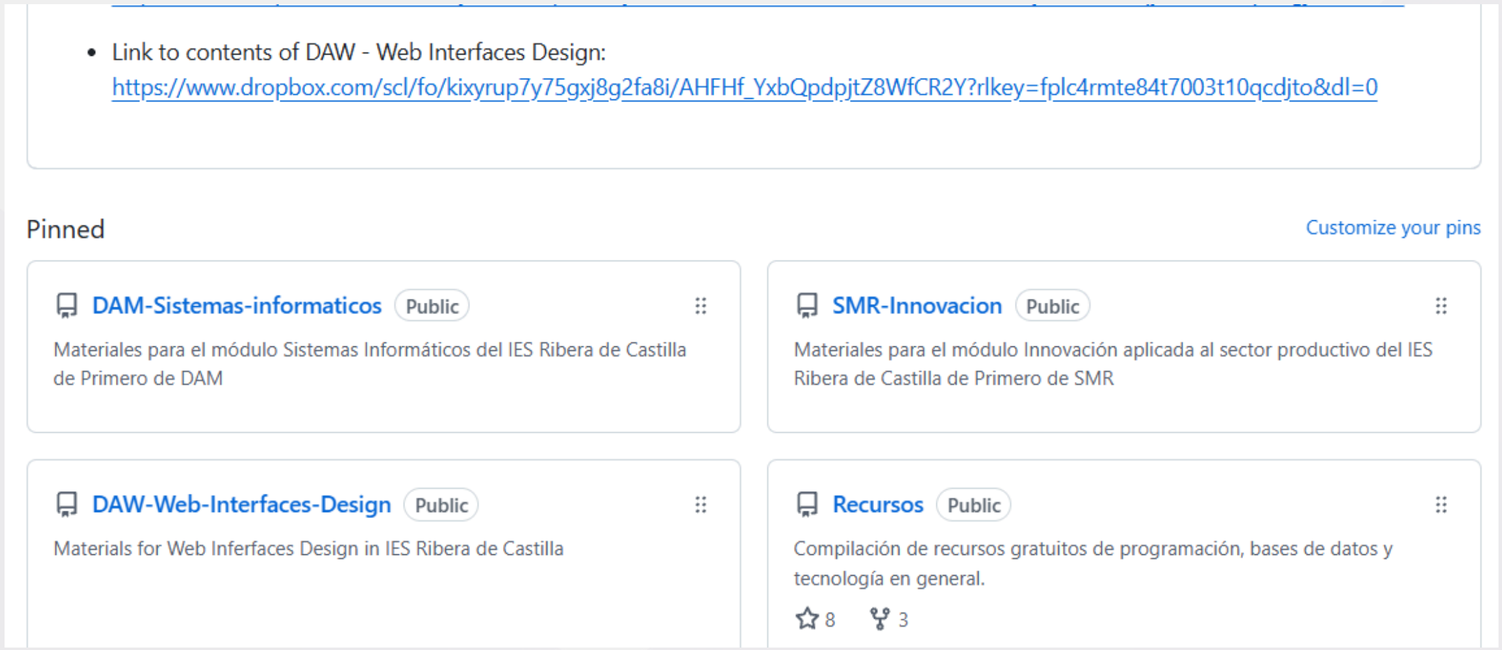
BOOK


INFORMÁTICA Y COMUNICACIONES
G.S. Administración de Sistemas Informáticos en Red
Fundamentos de hardware
Juan Carlos Moreno Pérez
EDITORIAL SINTESIS

MATERIALS

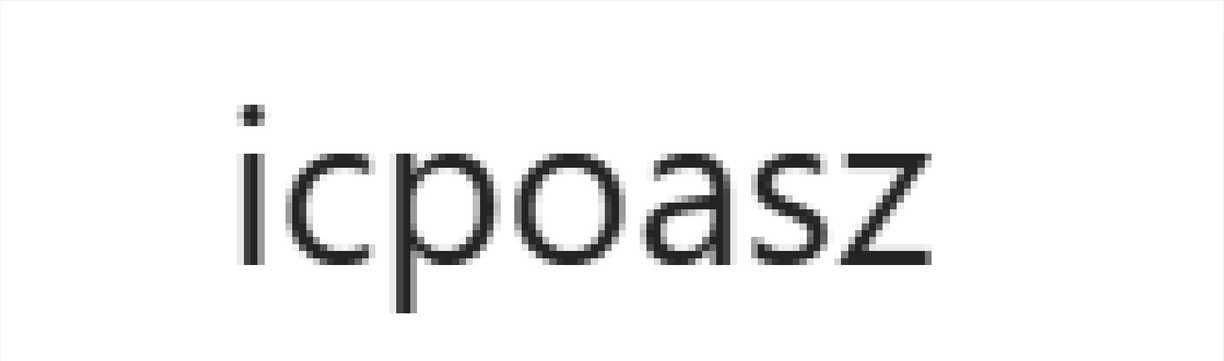
GitHub: https://github.com/BegoRodriguez/DAW-Web-Interfaces-Design
Dropbox: https://www.dropbox.com/scl/fo/kixyrup7y75gxj8g2fa8i/AHFHf_YxbQpdpjtZ8WfCR2Y?rlkey=fplc4rmte84t7003t10qcdjto&dl=0



TEAMS

I have created a classroom team for communication and project and activities delivery on Teams.

This is the code:



COMPUTER ARCHITECTURES: FUNCTIONAL ELEMENTS OF A COMPUTER

Objetives:

- Explore computing from the perspective of computer hardware and systems.
- Understand the classical Von Neumann architecture, including its functional units and subsystems.
- Learn about the architectures currently used in mobile devices.
- Understand the concept of System-on-Chip (SoC).
- Identify emerging processor architectures, including neuromorphic chips and AI-oriented architectures.

COMPUTER ARCHITECTURES: FUNCTIONAL ELEMENTS OF A COMPUTER (II)

- Introduction
- Computer Architecture. Functional Units and Subsystems
  - Memory
  - Input/Output Subsystem
  - The ALU (Arithmetic Logic Unit)
  - The Control Unit
- SoCs. Mobile Processors
- New Microprocessor Architectures: Neuromorphic Chips
- Artificial Intelligence in Processors

THINK ABOUT IT FOR A MOMENT...

Computers are part of everything we do. We use them without even noticing: surfing the web, writing, emailing, gaming, watching content. So let’s start with a simple but powerful question: what is a computer, really? An appliance? A toy? A typewriter? Something more?

AUTOEVALUATION

A computer is...

... a machine for writing documents
... a calculator
... a general purpose machine that works with given data and provides results
... a machine to see TV

AUTOEVALUATION (II)

A computer is...

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COMPUTER DEFINITION

A computer is an electronic, programmable device capable of receiving data as input, processing and storing it according to a set of instructions, and producing information as output.

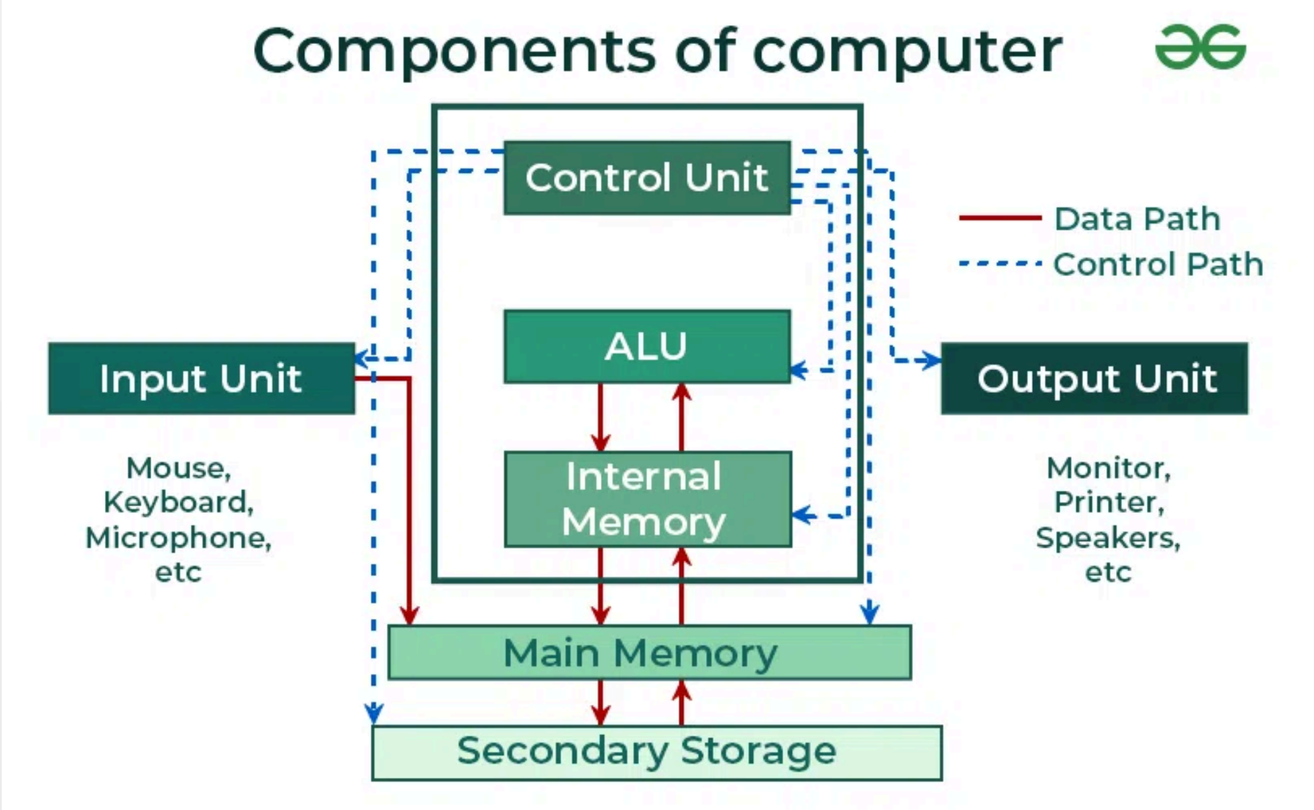
A computer system and is composed of two parts: hardware (the physical elements) and software (the instructions that tell the hardware what it has to do, how to solve the problem). Sometimes, some authors include two parts more: the user and the data.

ACTIVITY: COMPUTER HISTORY

Write a short article about the origins of computer science and the development of the first computers. Include references to early machines such as the Pascaline, Colossus, ENIAC, and the Intel 8086, among others.

COMPUTER ARCHITECTURE

Computer architecture is a theoretical model that describes the structure, components, and organization of a computer system



VON NEUMANN ARCHITECTURE

Von Neumann architecture is an abstract model created in 1946 that describes a computer architecture.

In Von Neumann architecture, the main elements of a computer are:

- CPU (Central Process Unit).
- Memory unit.
- Buses.
- I/O (Input/Output to external devices).

VON NEUMANN ARCHITECTURE (II)

Von Neumann architecture works as described:

- Components areconnected by buses.
- It can execute elemental instructions (machine code).
- The instructions are stored in the memory unit. Also, the data has to be stored in memory unit.
- Both instructions and data could be provided by input devices or generated by other programs (compiler, linker, etc.).
- The CPU reads an instruction and the control unit generates each signal to perform the instruction.
    - If it is required, CPU can perform arithmetic-logic operations.

- The CPU results can be stored on the memory unit, or it can be sent to output devices.

VON NEUMANN ARCHITECTURE (III)

Over the years, processors and memory have advanced dramatically to meet growing computational needs → Moore's law

Modern CPUs achieve remarkable performance, and memory systems provide massive storage capacity.

This evolution has led to highly specialized components, enabling microprocessors to integrate billions of transistors on a single chip.

Nowadays, there are several improvements of this architecture, including features like mapped I/O (memory unit and I/O devices are treated like the same memory).

If you want to try how Von Neumann architecture works, you can use this website: https://lab.xitrus.es/VonNeumann/

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HARVARD ARCHITECTURE

Harvard's architecture is other abstract model that describes a computer architecture. It is less implemented than Von Neumann architecture.

The main differences between Harvard architecture and Von Neumann architecture are:

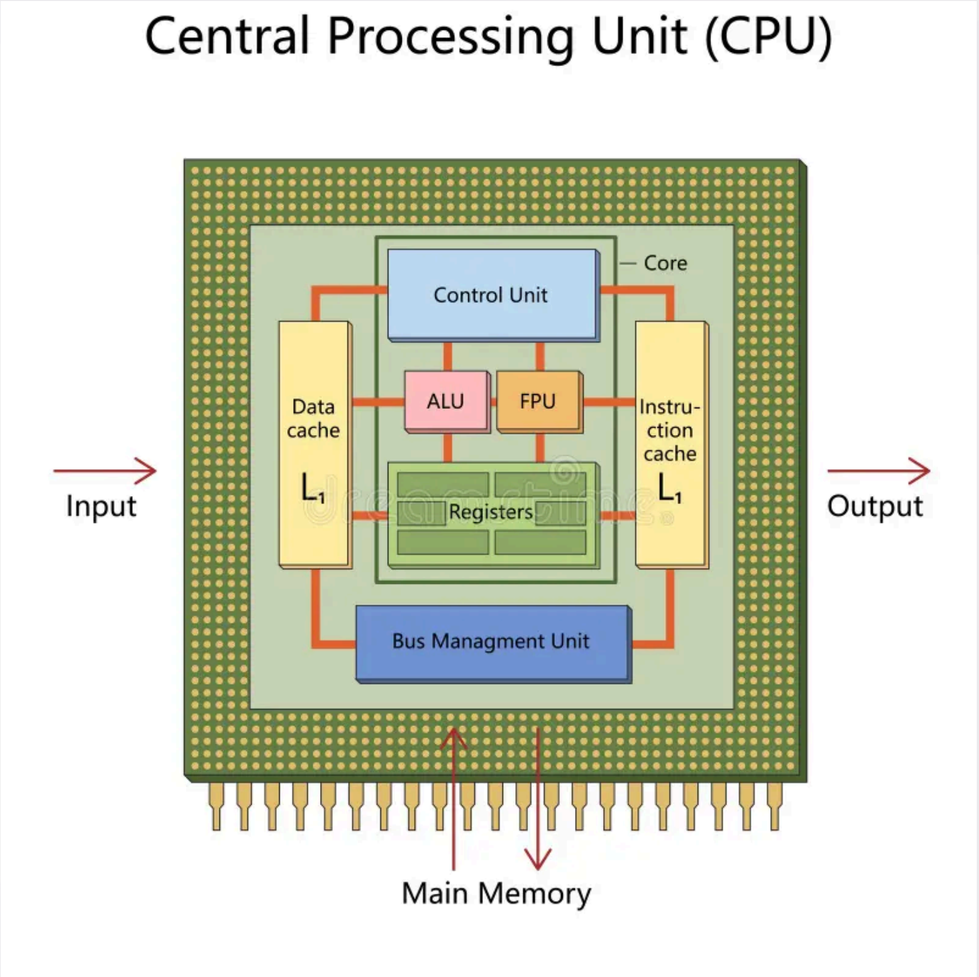
- In Von Neumann architecture, data and instructions are stored on the same memory. CPU only can read an instruction or a data at the same time. In Harvard architecture, data and instructions are stored separately, and each one has its own bus. CPU can read both at the same time (an instruction and a data).

- In Von Neumann architecture, data and instructions are stored on the same memory, and they share address space. In Harvard architecture, they are stored in different memories.

HARVARD ARCHITECTURE (II)

IMPORTANT

- Usually, real computer don’t use pure Von Neumann architecture or pure Harvard architecture. The only are abstract models.



FUNCTIONAL ELEMENTS OF A COMPUTER

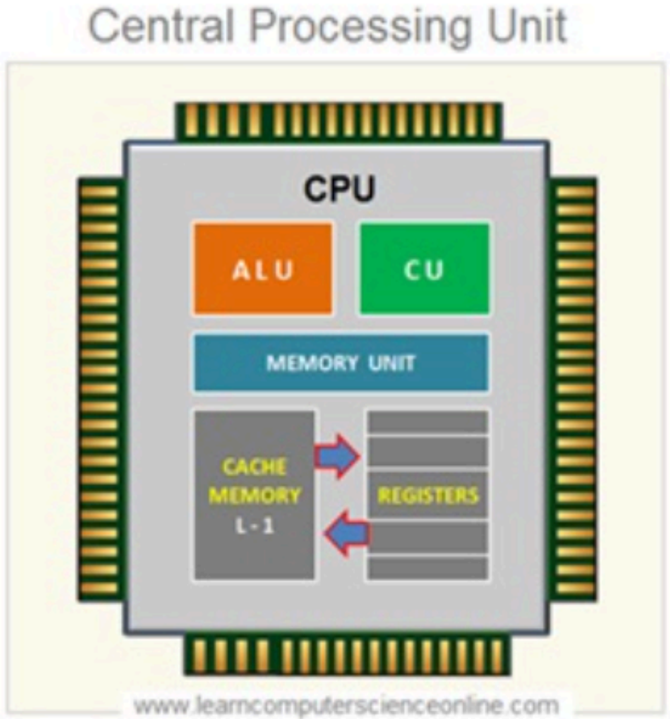
In this point, we are going to describe functional elements of a computer that implements Von Neumann architecture.

CPU(CENTRAL PROCESS UNIT)

The CPU is the core of the computer. CPU read instructions from memory and execute them, doing the required operations. It is composed of:

- **Registers:** they are little and fast memories. They are used as temporal storage when an instruction is being executed or to store temporally results of operations. The most important registers are:
    - **Program counter (PC):** an incremental counter that stores the memory address of the next instruction that is going to be executed.
    - **Memory address register (MAR):** it stores the address of a block of memory for reading from or writing to.

CPU REGISTERS



CPU Registers

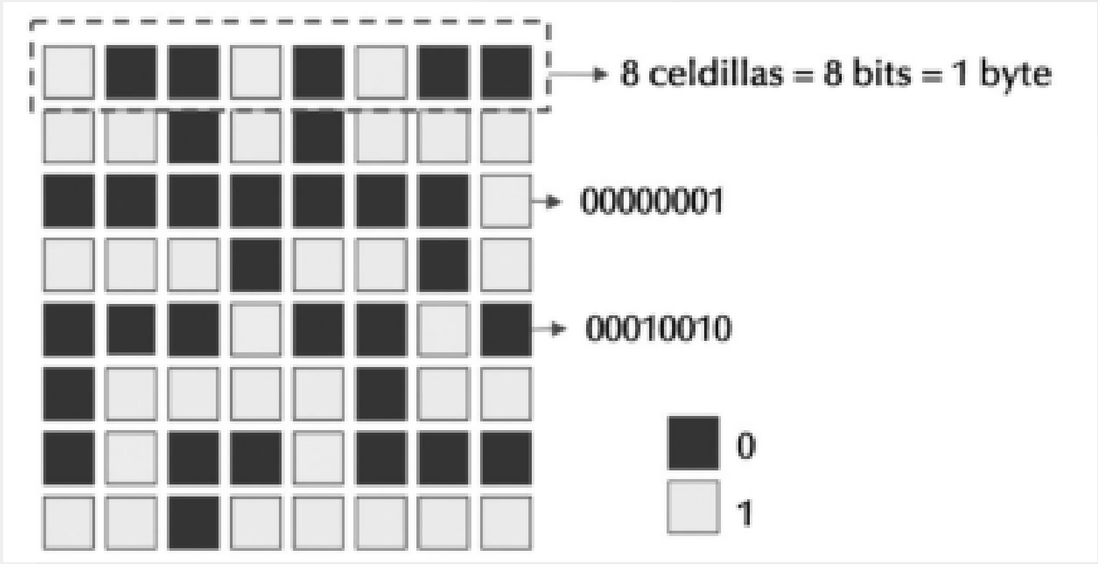
Special Purpose Registers

General Purpose Registers

The CPU Registers are High Speed memory area inside the processor chip And used by the processor during the program execution.

CPU REGISTERS (II)

- The memory inside the CPU (registers) is made up of a series of **cells** or **bistables** (so called because they can only hold two possible states)
- Each one stores a single bit.
- Access to a group of these cells is generally performed through an address.
- These cells can be accessed in both read and write modes.



CPU REGISTERS (III)

- The cells are the basic information units, called bits. These bits can take two possible values, 1 and 0 (presence or absence of electrical energy).

- These cells are used to store the machine’s data and instructions. These instructions are grouped into what are known as programs. Programs are sets of instructions stored sequentially, and their instructions are executed one after another. The system executes these programs, which are located in main memory.

- The data used by the programs, as mentioned, also reside in main memory, and the system must fetch the instructions, interpret them, and access the data whenever necessary, either in write or read mode.

CPU (CENTRAL PROCESS UNIT) (II)

- **Registers (II):**
  - **Memory data register (MDR):** a two-way register that stores data fetched from memory (and ready for the CPU to process) or data waiting to be stored in memory.
  - **Instruction register (IR):** it is a temporal register to store the instruction that has just been fetched from memory.
  - **General purpose registers:** those registers are used to store information temporally, for example to store operands of an addition or to store the result of an arithmetic-logic operation.
  - **Flags:** These are very simple records. They can only be in two states: “On” or “Off”. They are used to indicate the status of an operation. Example: flag Z is set to 1 if the operation has resulted in 0, N is set to 1 if the result is negative.

CPU (CENTRAL PROCESS UNIT) (III)

- **Internal bus:** is an internal bus that connect registers, control unit and arithmetic-logic unit.
- **Control unit (UC):** it decides what to do in each moment. Its components are:
  - **Decoder:** its function is to decode instructions and prepare all the signals to execute the program properly.
    - For instance, an instruction can be “ADD” (which adds up two numbers that are in memory). When it comes to executing, the decoder generates a lot of small signals such as reading the first number of the memory, taking it to the ULA, reading the second number, taking it to the ULA, perform the operation (adding) and saving the result in memory. In this case, an instruction has generated six signals 1.
  - **Clock:** is the system that sets the pace at which things are done. It's like the conductor's baton. It allows all system components to operate perfectly together.

CPU (CENTRAL PROCESS UNIT) (IV)

- **Arithmetic-Logic Unit (ULA):** it is a unit designed to perform arithmetic operations (addition, subtraction, multiplication, etc.) and logic operations (not, and, or, ...). It's the worker.

To execute an instruction, the CPU follow those steps:

- Use the Program Counter to fetch the new instruction from memory.
- Decode the instruction and fetch data from memory if it is required.
- Perform the instruction and store data in an internal register or memory.

The instruction cycle will be detailed in later in the section Instruction cycle.

to measure the speed of a CPU, we use frequency units that indicate the number of times a thing repeats itself per unit of time. In general, we use the Hz (or its multiples) that indicates the number of operations that can be performed in a second.

CPU (CENTRAL PROCESS UNIT) (V)

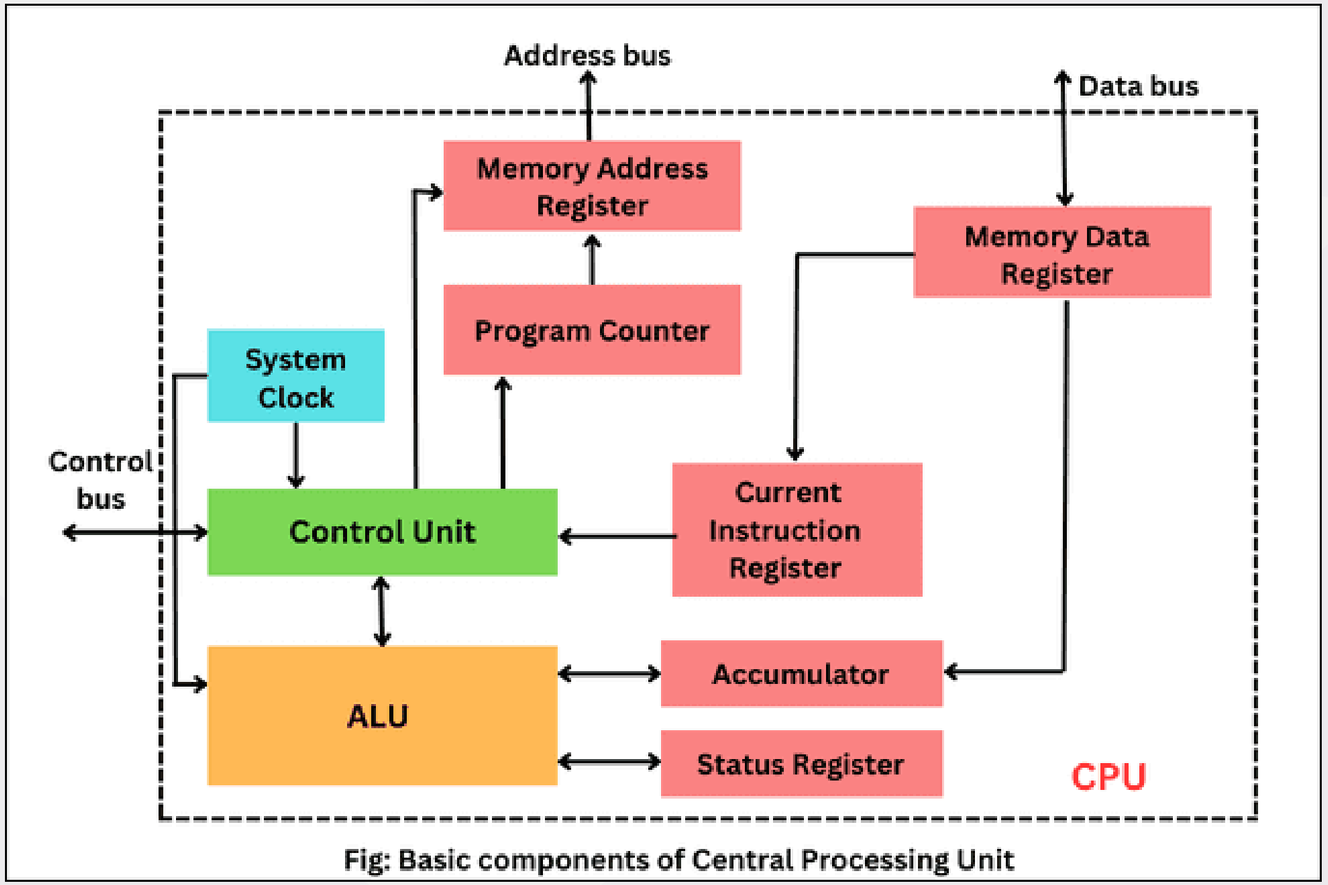
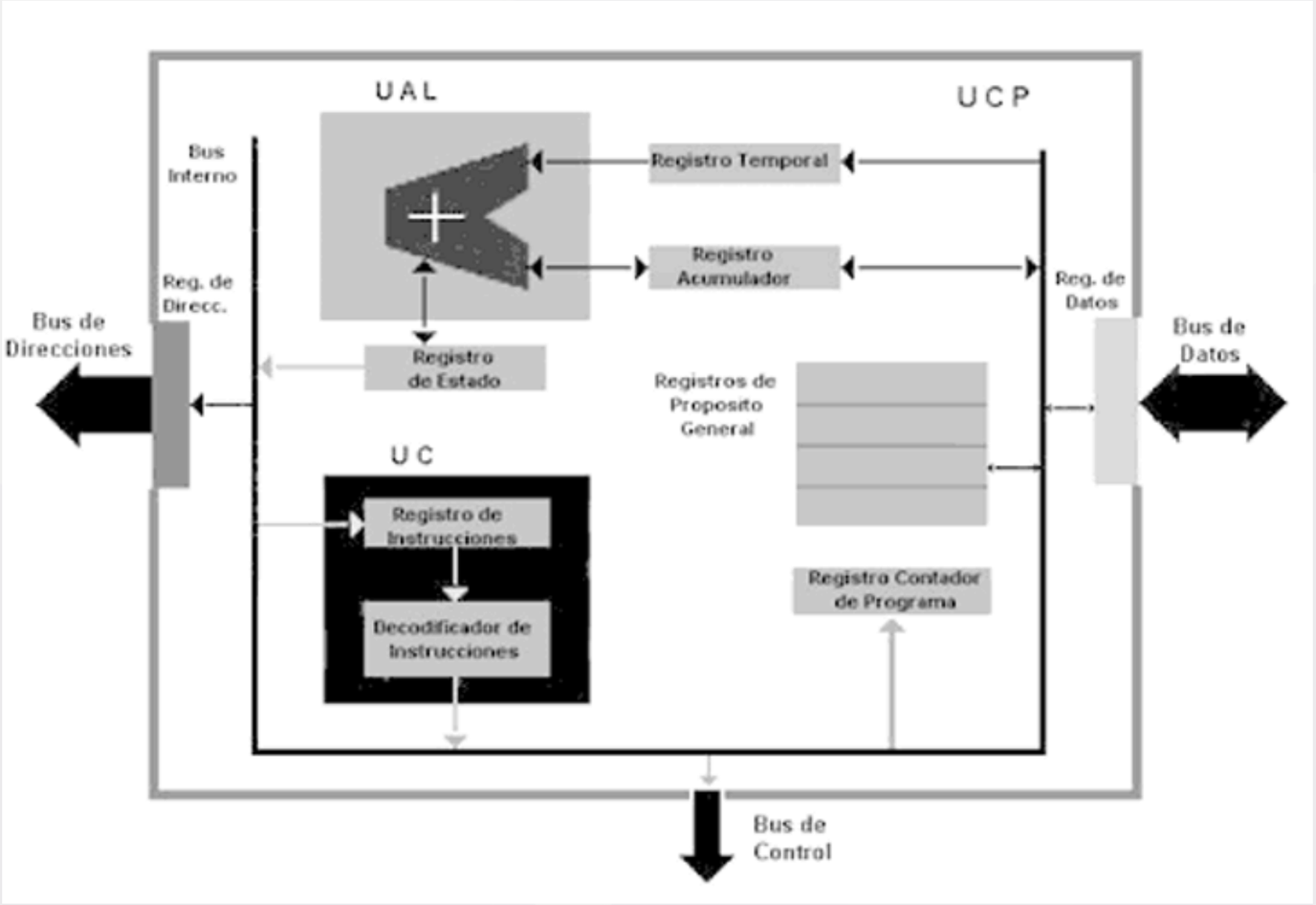
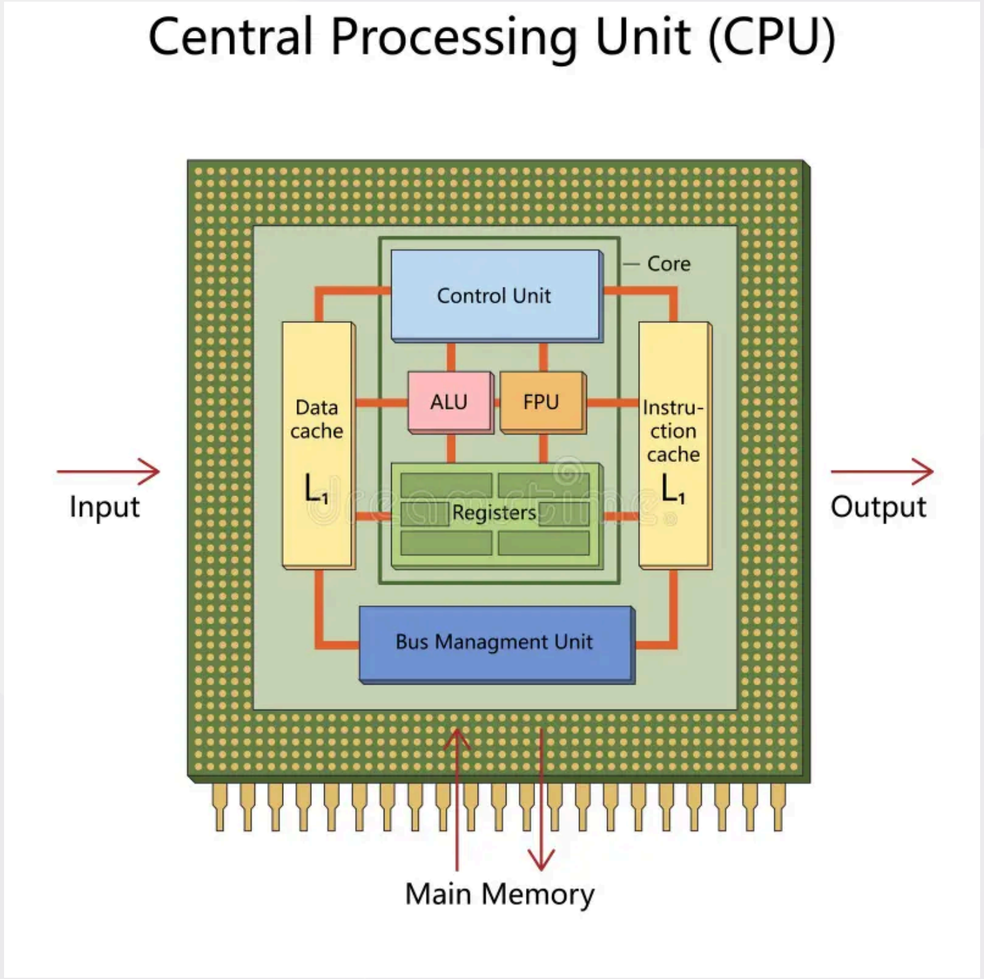


Fig: Basic components of Central Processing Unit

CPU(CENTRAL PROCESS UNIT) (VI)



CPU(CENTRAL PROCESS UNIT) (VII)

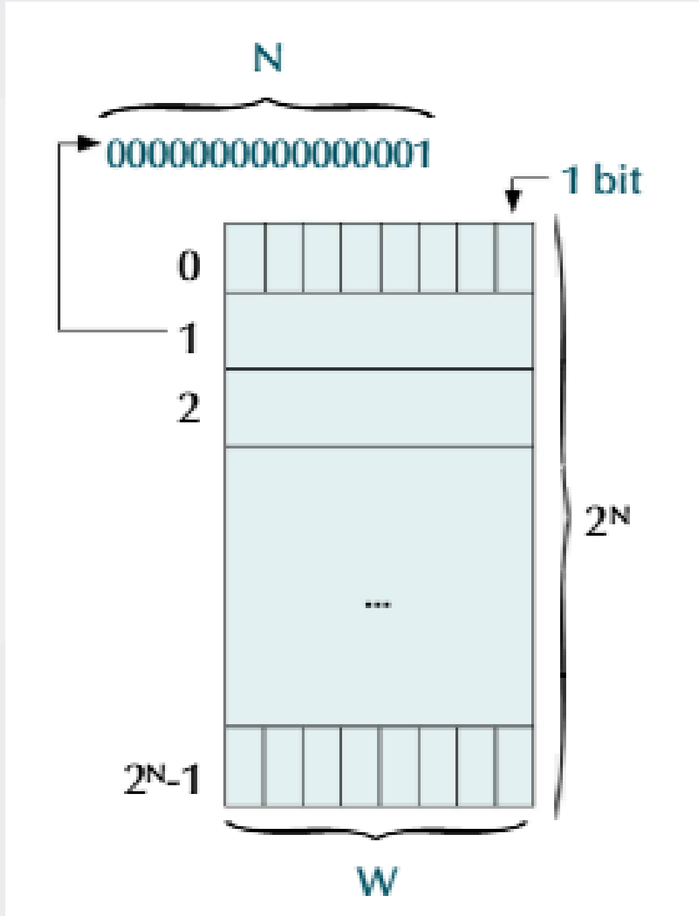


RAM MEMORY: BASIC STRUCTURE

- RAM (Random Access Memory) consists of a series of memory cells, each containing several bits.
- Each bit stored in a cell can hold one of the following values: 0, 1, or no value.
- Typically, each **cell** contains one **byte** (1 byte = 8 bits).
- RAM is used as temporary storage, allowing fast access to data needed by the CPU.
- All these cells have a specific address, and the cell is the minimum unit of memory, which can be fully read or written.

RAM MEMORY: BASIC STRUCTURE (II)

- Memory consists of cells numbered 0, 1, …, 2^N −1, each with a binary address.
- If the system uses two bytes (16 bits) to address memory, cell 1 is represented as 0000000000000001, and accessing that cell requires specifying its full binary address.
- It must also be considered that the number of bits used to address a memory cell directly determines the **total memory capacity** of the system.



RAM MEMORY: BASIC STRUCTURE (III)

It is important to distinguish between a memory cell’s address and its content:

- **Address of a memory cell:** The address is expressed as a binary number (typically two bytes) and indicates where the data is located in memory.

- **Content of a memory cell:** The address points to a cell that contains a value (in this example, 1 byte = 8 bits).

PROGRAM EXECUTION AND MEMORY ACCESS

- The time the system takes to read or write any memory cell is the same, regardless of its position.
- Programs consist of a sequence of instructions executed by the processor, and to be executed, they must reside in memory.
- The data used by these programs must also be stored in memory so the CPU can access them when needed.
- **Memory access time** is the amount of time required to perform a read or write operation, typically measured in nanoseconds — where one nanosecond is one billionth of a second.

MEMORY OPERATIONS: READ AND WRITE

a) Read (address):

- Reading from a memory address means retrieving the content stored in a specific memory cell.
- The read operation does not modify the value of the cell; the data remains unchanged.
- It is a non-destructive operation, so it can be repeated without altering the stored information.

MEMORY OPERATIONS: READ AND WRITE (II)

b) Write (address, value):

- Writing to a memory address means storing a value into a specific memory cell.
- The write operation modifies the cell’s content by copying data from a register into that memory location.
- It is a destructive operation in the sense that it replaces the previous value with a new one.

MEMORY OPERATIONS: READ AND WRITE (III)

To perform memory operations, the system must send either a read signal or a write signal to the memory module.

Two essential registers are required for these operations: MAR (Memory Address Register) and MDR (Memory Data Register)

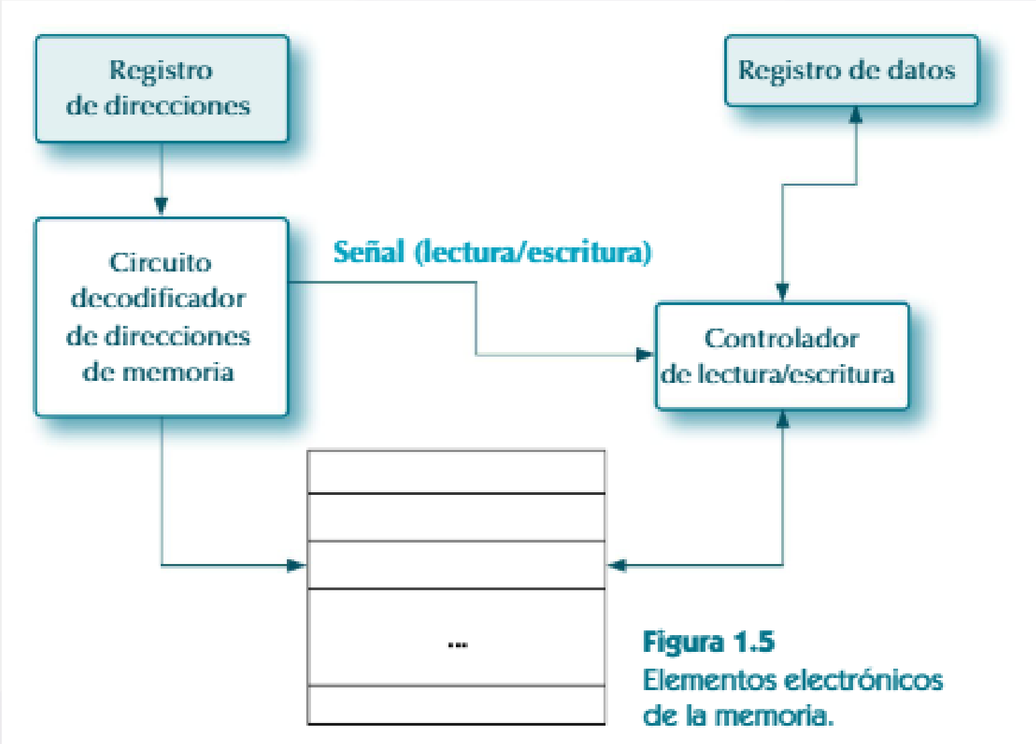
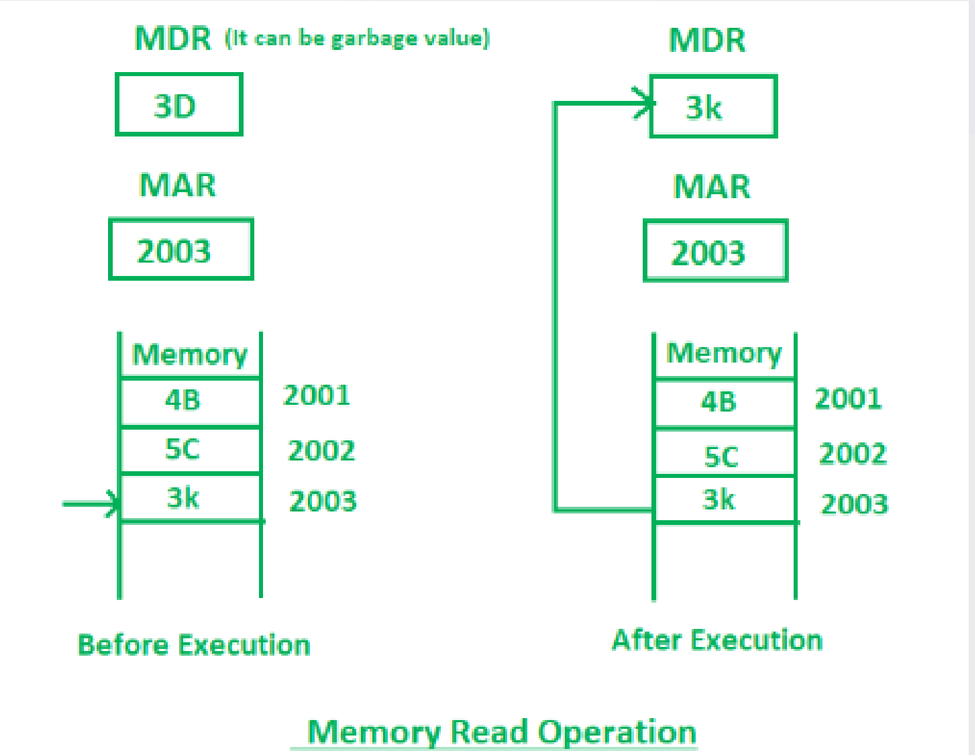


Figura 1.5
Elementos electrónicos de la memoria.

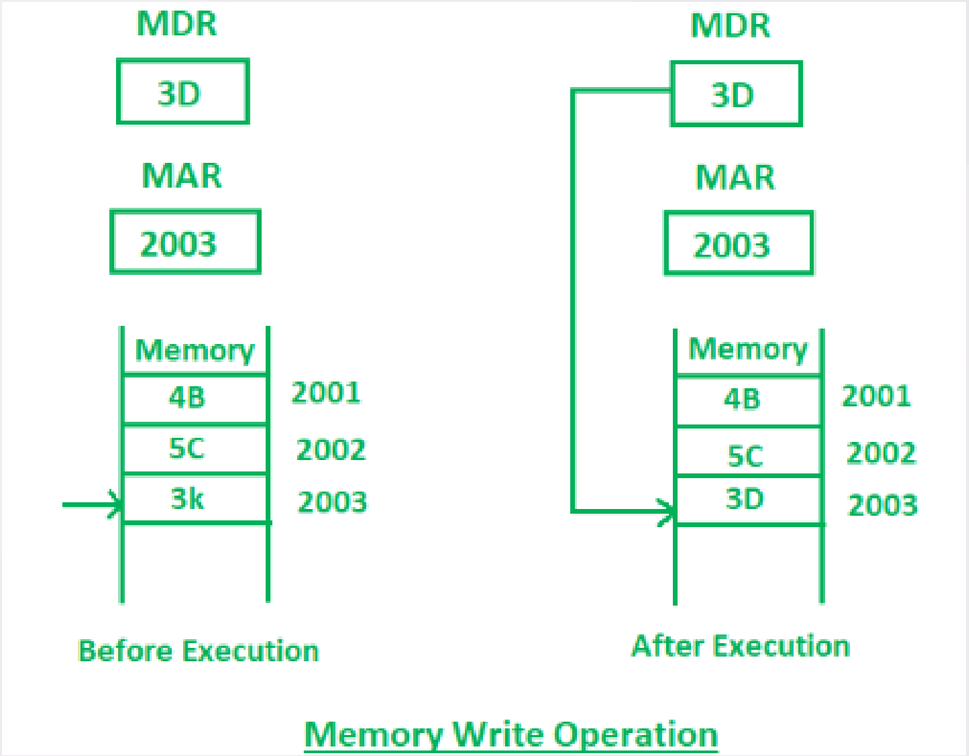
MEMORY OPERATIONS: READ

- The address is loaded into the address register.
- The address is decoded, locating the specific memory cell to be accessed.
- The content of that memory cell is copied into the data register.
- The value stored in memory remains unchanged, since reading is a non-destructive operation.



MEMORY OPERATIONS: WRITE

- The address is loaded into the address register.
- The value to be stored is placed in the data register.
- The address is decoded, locating the specific memory cell where the data will be written.
- The content of the memory cell is replaced with the value from the data register.
- Writing is a destructive operation, meaning the previous value in that cell is overwritten.



Memory Write Operation

INPUT/OUTPUT SUBSYSTEM

- The input/output (I/O) subsystem controls the devices that allow the computer system to communicate with the external environment.
- These devices include: Display (monitor), Keyboard, Printer, Mouse and other peripherals that enable user interaction.
- The I/O subsystem also manages secondary storage devices, such as hard drives, solid-state drives, or external storage units, allowing the system to store and retrieve information beyond main memory.
- It acts as a bridge between the CPU and the outside world, ensuring smooth data exchange and device coordination.

MEMORY AND STORAGE ACCESS TIME

- The time required for memory to perform a read or write operation is usually measured in nanoseconds (ns).

- In contrast, hard drives and other secondary storage devices measure access time in milliseconds (ms).

- This difference shows that main memory (RAM) is millions of times faster than secondary storage, which is why data must be loaded into memory before the CPU can process it efficiently.

- Example comparison:
  - RAM: ~10–100 nanoseconds
  - Hard drive: ~5–10 milliseconds
  - SSD: ~0.1–1 millisecond

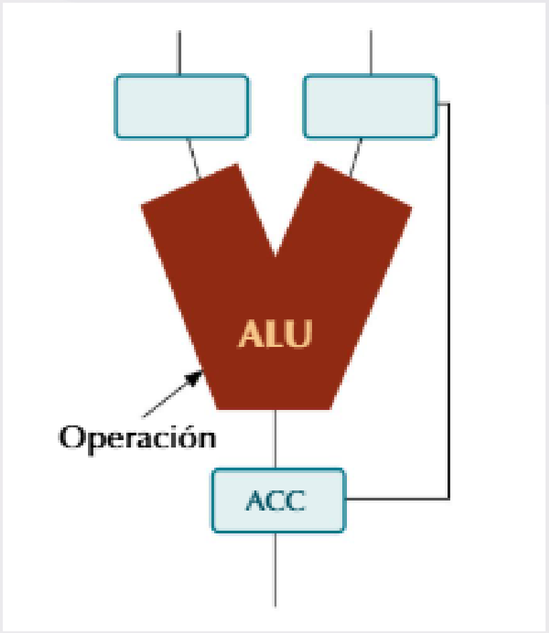
THE ALU (ARITHMETIC LOGIC UNIT)

- It is an integrated circuit responsible for performing arithmetic operations (+, −, ×, ÷) and logical operations (=, <, >, AND, OR, NOT...).

- It consists of:
  - A set of circuits that execute arithmetic and logical operations.
  - A series of registers that store intermediate results during computation.
  - A bus that interconnects the registers with the operation circuitry.

- The ALU can take up to two operands at a time.

- The result of the operation is typically stored in a special register called the accumulator (ACC).



THE CONTROL UNIT (CU)

- It is responsible for executing program instructions, one by one.
- Programs are stored in main memory as binary machine instructions, the only language the hardware understands.
- The CU fetches each instruction from memory, interprets it, and coordinates the actions of the CPU components (ALU, registers, buses...).
- It ensures that every step of the instruction cycle (fetch → decode → execute) happens in the correct order.
- The CU acts as the “conductor” of the computer system, directing data flow and timing so operations occur smoothly and correctly.

CPU FETCH-EXECUTE CYCLE

1. **Fetch** the instruction from memory

   - The CU reads the next instruction stored in main memory.
   - The address of the instruction is typically held in the Program Counter (PC).

2. **Decode** the instruction

   - The CU interprets the binary pattern to determine which operation must be performed.
   - It identifies the required operands, registers, and functional units.

CPU FETCH-EXECUTE CYCLE (II)

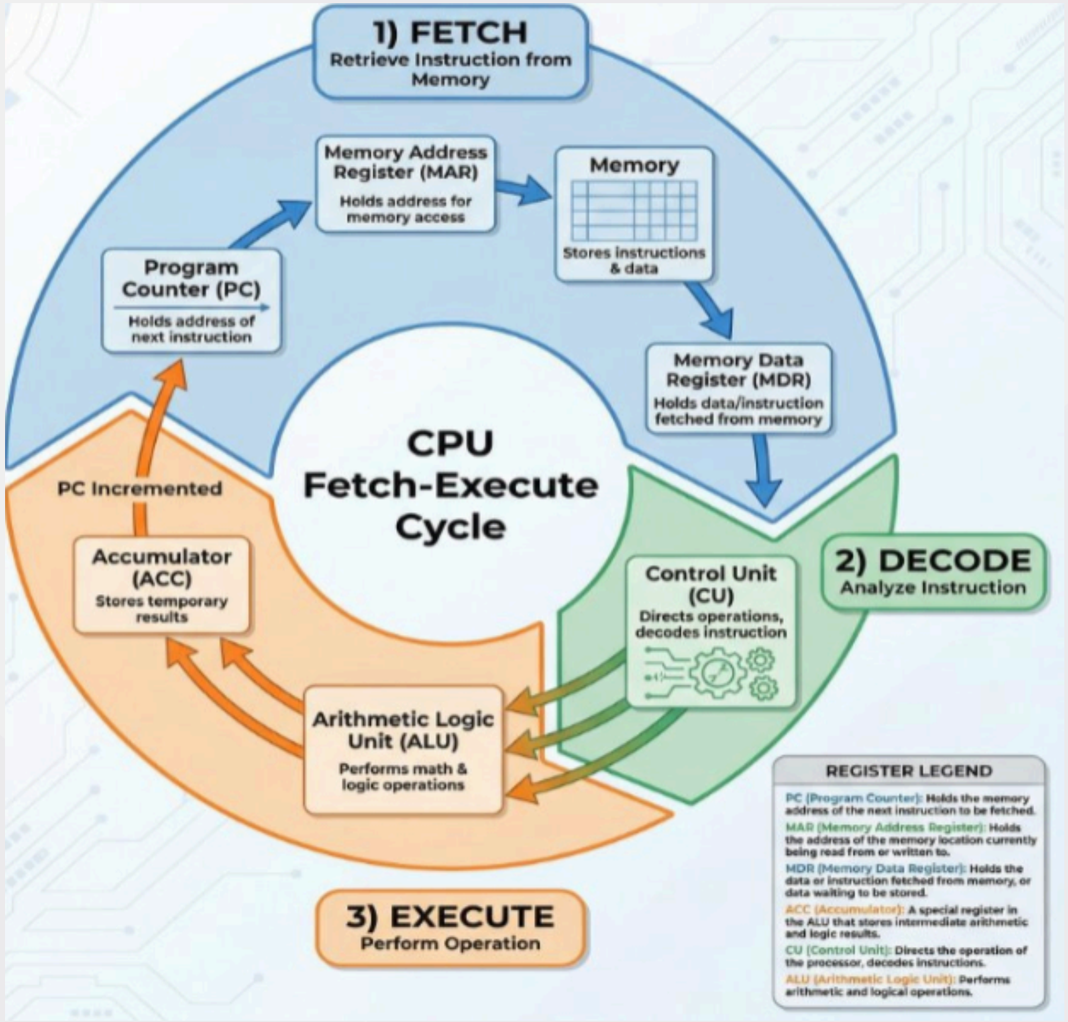
3. **Execute** the instruction

- The CU sends the appropriate control signals to:
  - ALU (for arithmetic/logic operations)
  - Memory (for read/write operations)
  - I/O subsystem (for input/output operations)
- Each component performs its part of the instruction.

4. **Continue** execution until HALT

- The CU repeats the cycle (fetch → decode → execute) for the next instruction.
- The process continues until a HALT instruction is encountered, signaling the end of the program.

CPU FETCH-EXECUTE CYCLE (III)



STRUCTURE OF A MACHINE INSTRUCTION

- A machine-language instruction is composed of two main fields:
  - Operation code (**opcode**)
  - Values or memory addresses containing the operands needed for the operation

- Example: “SUM A + B”
  - Suppose the instruction is SUM A + B.
  - The opcode for the SUM operation is 105, which in binary is 1101001.
  - The operands A and B correspond to the values 7291 and 15484, which also are represented in binary.

PRACTICAL CASE: OUR HW1 MACHINE

1. Create a program using HW1 machine instructions that reads two values from the keyboard and displays their sum on the screen

```
READ RA
READ RB                 // We read the numbers to add in registers A and B
MOVE ACC RB             // We move into ACC operand B
SUM ACC RA              // We sum ACC that contains B with RA that contains A
MOVE RC ACC             // We take the result to register C
PRINT “THE SUM IS:”
PRINT RC                // We print the result on the screen
```

PRACTICAL CASE: OUR HW1 MACHINE

2. This time, create a program using HW1 machine instructions that reads three values from the keyboard and displays their sum on the screen.

```
READ RA
READ RB
READ RC                 // We read the numbers to add in registers A, B and C
MOVE ACC RB             // We move into ACC operand B
SUM ACC RA              // We sum ACC that contains B with RA that contains A
SUM ACC RC              // We sum ACC that contains A+B with RC that contains C
MOVE RD ACC             // We take the result to register D
PRINT “THE SUM IS:”
PRINT RD                // We print the result on the screen
```

PRACTICAL CASE: OUR HW1 MACHINE

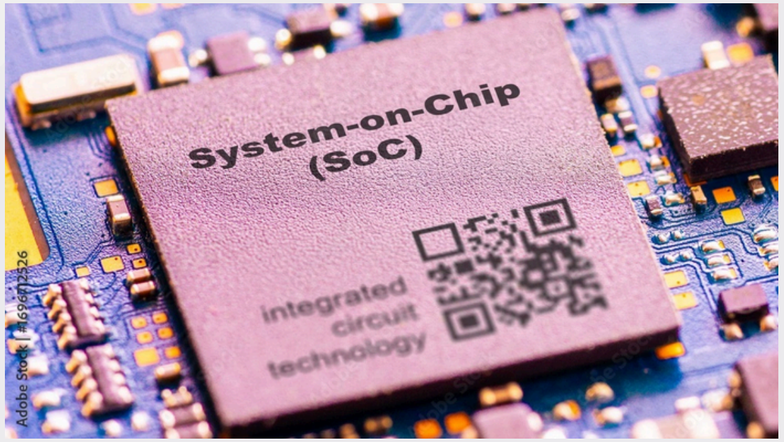
3. Create a program using HW1 machine instructions that displays the numbers from 1 to 10 on the screen.

```
MOVE ACC 11        // A counter stored in A, when that reachs 11 program ends
MOVE RB ACC
MOVE ACC 1
MOVE RA ACC                  // We initialize RA with 1
LOOP:                        // INTX
PRINT RA
MOVE ACC 1
SUM ACC RA
MOVE RA ACC
MOVE ACC RB
SUB RA ACC                   // We substract 11. If this is zero program must end
JUMP IF NOT ZERO LOOP  // It could also be 5
PRINT “End of program”
```

SOCS. MOBILE PROCESSORS

Modern devices, together with their computing power and cloud platforms, have turned **mobile computing** into a new way of working. One of the decisive factors that has made mobile computing a reality is the power of the new microprocessors (SoCs) designed for small devices.

These microprocessors are called **System on a Chip (SoC)** because the processor itself performs functions that, in a traditional architecture, would be handled by other components.



With technological evolution, SoCs have gained increasing processing power and can carry out more complex functions.

SOCS. MOBILE PROCESSORS (II)

Since 1978, microprocessors have followed the x86 architecture. However, with the rise of smartphones and tablets, the **ARM architecture** has become highly significant, as these devices are based on it.

Initially, this architecture was designed for devices with very low processing capacity and low power consumption, unlike the x86 architecture, which prioritizes computing power and speed.

This evolution has made it possible to design microprocessors that, in addition to data processing, also handle other features such as Bluetooth, input/output, graphics processing, and more.

ARM VS X86 - KEY DIFFERENCES

Aspect	ARM Architecture	x86 Architecture
Instruction Set	RISC (Reduced Instruction Set Computing) — simple, fewer instructions	CISC (Complex Instruction Set Computing) — large, complex instruction set
Energy Consumption	Very low; optimized for efficiency and battery-powered devices	Higher; designed for performance over efficiency
Performance Focus	Efficiency, multitasking, mobile workloads	Raw performance, high-speed computation
Typical Devices	Smartphones, tablets, IoT, embedded systems	PCs, laptops, workstations, servers
Integration Level	Often part of a **SoC** (CPU + GPU + I/O + connectivity)	Components usually separated (CPU + chipset + GPU)
Heat Generation	Low	Higher
Power Source	Battery-oriented	Plugged-in systems
Best Use Cases	Mobile computing, low-power environments	Heavy computation, desktop/server workloads

ARCHITECTURE CLASH: CISC VS RISC

Aspect	CISC (Complex Instruction Set Computing)	RISC (Reduced Instruction Set Computing)
Instruction Set	Large and complex; many powerful instructions	Small and simple; fewer, optimized instructions
Instruction Complexity	Instructions can perform multiple operations	Each instruction performs a single, simple operation
Execution Time	Often slower per instruction due to complexity	Faster per instruction; simpler operations
Hardware Requirements	More complex hardware; larger control unit	Simpler hardware; smaller control unit
Code Size	Typically smaller (fewer instructions needed)	Typically larger (more instructions needed)
Energy Consumption	Higher	Lower; ideal for mobile and embedded devices
Performance Focus	Raw computational power	Efficiency and speed through simplicity
Typical Architectures	x86 (Intel, AMD)	ARM (mobile devices, IoT, embedded systems)
Best Use Cases	Desktops, laptops, servers	Smartphones, tablets, low-power systems

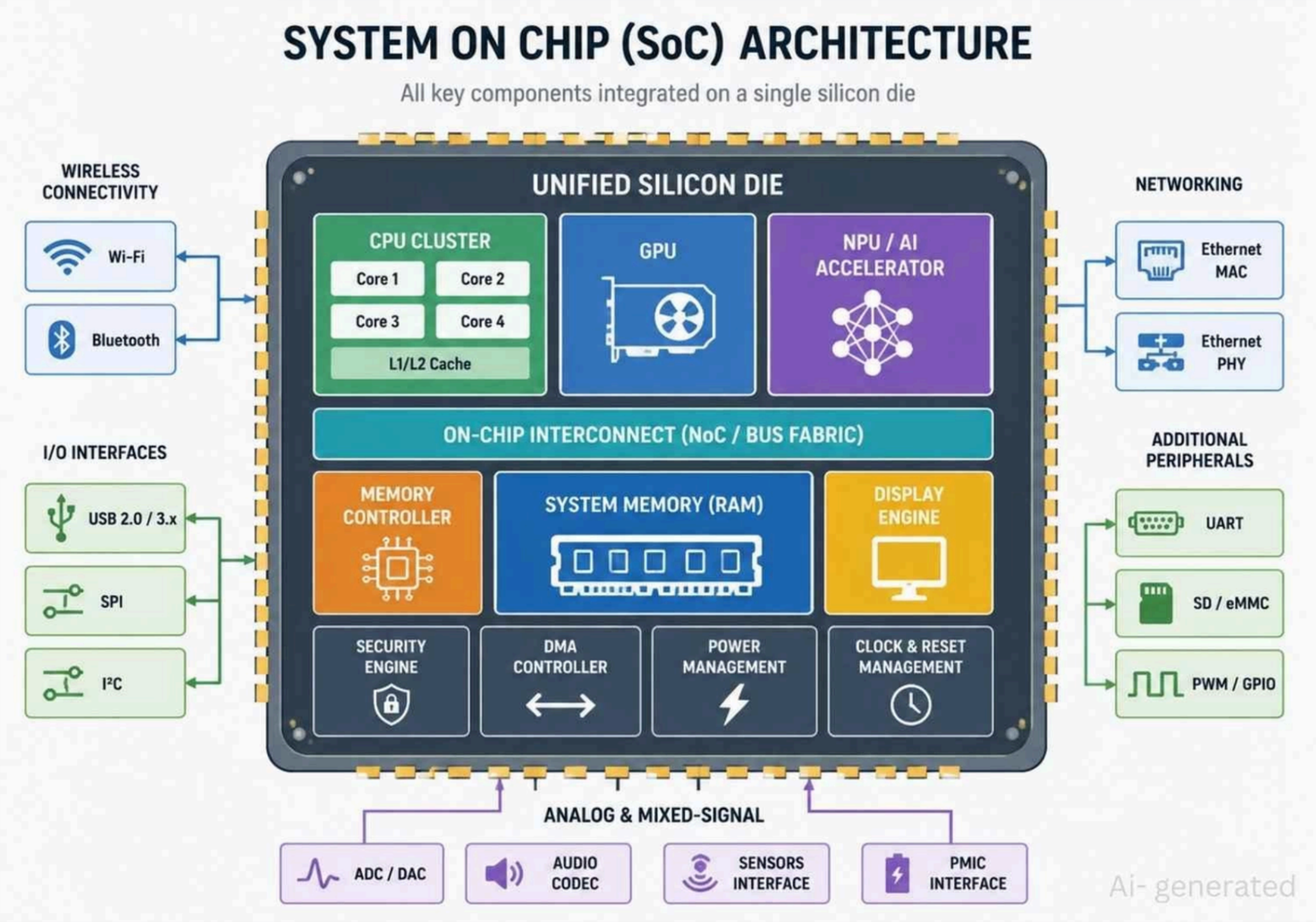
SOCS. MOBILE PROCESSORS (III)

The speed of a SoC depends on the number of **cores**.

A **core** in a microprocessor is the fundamental execution unit inside a CPU, capable of independently fetching and running program instructions. It’s essentially a mini-processor within the larger processor package.

Apart from how efficient it is, the more cores a processor has, the more tasks it can execute simultaneously, and the greater its computing power will be. In addition to the number of cores, their clock speed must also be taken into account.

SOCS. MOBILE PROCESSORS (IV)



SOC COPROCESSORS

- Specialized units that assist the main processor.
- Originally used for complex mathematical calculations (FPUs, **F**loating-**P**oint **U**nit).
- Modern smartphones include graphics coprocessors (GPUs, **G**raphics **P**rocessing **U**nits) for photo and video processing.
- Improve performance and reduce power consumption by handling tasks more efficiently than the main CPU.

MOTION COPROCESSORS

- Apple's MX coprocessors work alongside the AX processor.
- They process data from sensors such as the compass, gyroscope, and accelerometer.
- By handling these tasks, they reduce the workload of the main CPU.
- Improve system performance and lower power consumption.
- Can detect when the device is idle and reduce system activity.
- Help extend battery life by operating more efficiently than the main processor.

GPU (GRAPHICS PROCESSING UNIT)

- One of the key components that determines the performance of a SoC.
- A specialized coprocessor designed for **graphics processing** and **floating-point calculations**.
- Offloads graphics-related tasks from the CPU, improving overall system performance.
- Provides the graphical power needed for applications, games, and multimedia content.
- Modern GPUs offer significant computing power but **cannot replace the CPU**, as both are designed for different purposes.
- A more advanced GPU can handle higher-resolution displays more efficiently while maintaining better performance and energy efficiency.

MANUFACTURING PROCESS (LITHOGRAPHY)

- One of the key factors that determines the advancement of a SoC.
- Refers to the size of the transistors and logic gates used to build the chip.
- Measured in nanometers (nm).
- Smaller manufacturing processes allow more transistors to fit into the same area.
- This results in better performance, lower power consumption, and greater energy efficiency.
- For example, a 14 nm process means that the chip's logic gates are approximately 14 nanometers in size.

PROPOSED ACTIVITIES

1. Develop a comparison between desktop processors and SoCs, highlighting their main characteristics, advantages, and disadvantages.

2. Reflect on the future of SoC technology and answer the following question: In which devices or applications do you think SoCs will be integrated in the future? Why?

3. Investigate the technological innovation introduced by Apple in its A11 chip and subsequent generations that made these processors more powerful and versatile.

PROPOSED ACTIVITIES (II)

4. Research and compare the following computing technologies:
   - Google Cloud TPU
   - NVIDIA GPUs used in high-performance computers

Consider their architecture, performance, advantages, and applications. Although NVIDIA is currently one of the leading companies in high-performance computing, Google has also developed its own engineering solutions with innovative products such as the Cloud TPU.

5. Consider the following machine instruction:

Operation Code (8 bits) Address 1 (8 bits) Address 2 (8 bits)

Assuming that each memory cell stores 8 bits, what is the maximum memory capacity of this machine?

PROPOSED ACTIVITIES (III)

6. Would it be possible to eliminate the main memory (RAM) from a computer system and operate using only the hard disk drive? Explain your answer.

7. Why is the ALU accumulator connected to the ALU buffer?

8. What is the difference between the ARM and x86 architectures? Compare their main characteristics and typical applications.

¿ANY QUESTION?